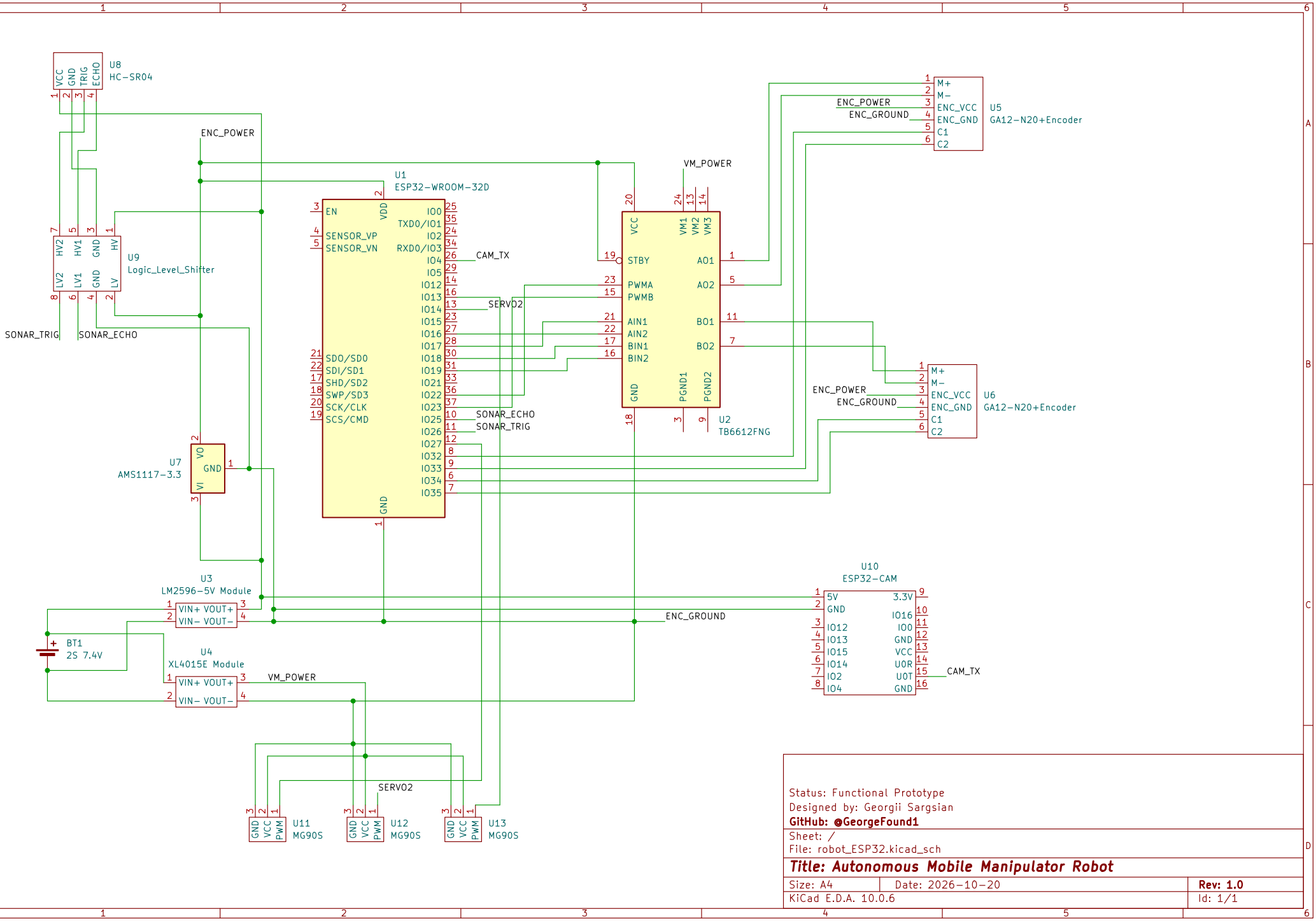

U8
HC-SR04
VCC
GND
TRIG
ECHO
ENC_POWER
U1
ESP32-WROOM-32D
EN
VDD
SENSOR_VP
SENSOR_VN
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
GND
CAM_TX
SERVO2
SONAR_ECHO
SONAR_TRIG
VM_POWER
U2
TB6612FNG
VCC
STBY
PWMA
PWMB
AIN1
AIN2
BIN1
BIN2
AO1
AO2
BO1
BO2
VM1
VM2
VM3
PGND1
PGND2
U5
GA12-N20+Encoder
M+
M-
ENC_VCC
ENC_GND
C1
C2
ENC_GROUND
U6
GA12-N20+Encoder
U9
Logic_Level_Shifter
HV2
HV1
GND
HV
LV2
LV1
LV
SONAR_TRIG
SONAR_ECHO
U7
AMS1117-3.3
VO
VI
U3
LM2596-5V Module
VIN+ VOUT+
VIN- VOUT-
U4
XL4015E Module
BT1
2S 7.4V
U10
ESP32-CAM
5V
3.3V
IO16
U0R
U0T
U11
MG90S
U12
MG90S
U13
MG90S
GND
VCC
PWM
Status: Functional Prototype
Designed by: Georgii Sargsian
GitHub: @GeorgeFound1
Sheet: /
File: robot_ESP32.kicad_sch
Title: Autonomous Mobile Manipulator Robot
Size: A4
Date: 2026-10-20
Rev: 1.0
KiCad E.D.A. 10.0.6
Id: 1/1